

Bakery Inventory & Sales Optimization Analysis

BALANCING PRODUCT AVAILABILITY AND INVENTORY EFFICIENCY

SUPPLY CHAIN ANALYTICS PORTFOLIO | ARIANNE ZELDA

Maintaining Inventory Availability While Minimizing Overstock Remains a Key Operational Challenge for Bakery Businesses

Background

Bakery operations require effective inventory management to ensure product availability while controlling inventory holding costs and minimizing stockout risk.

Objectives:

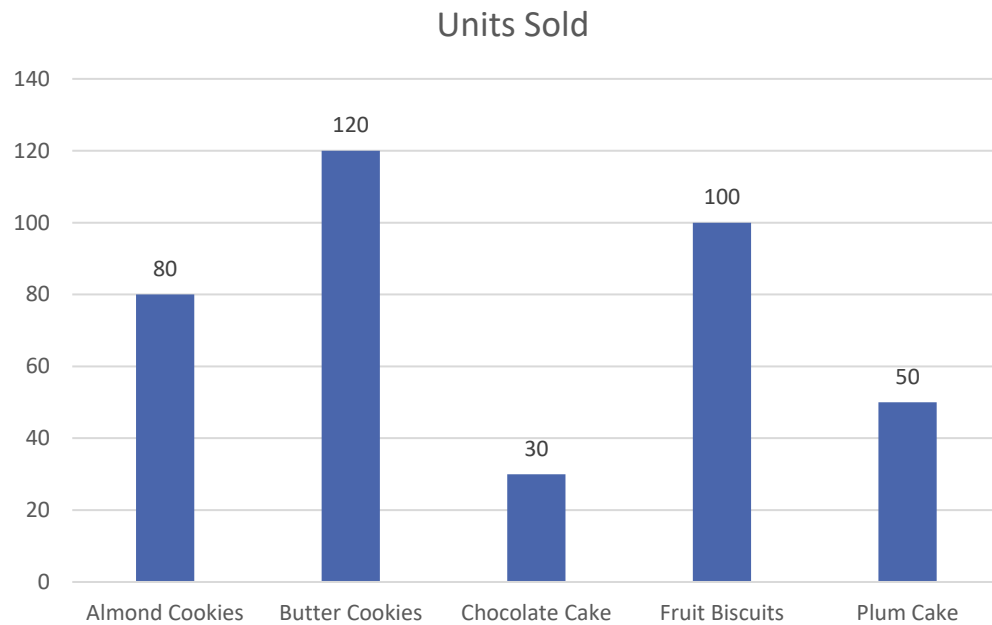
- Identify products with potential stockout risk
- Evaluate inventory efficiency across products
- Assess sales and demand performance
- Provide inventory replenishment recommendation

The Analysis Covers Inventory, Supplier, Sales, and Financial Data Across Five Bakery Products

Category	Variables
Inventory	Current Stock, Reorder Point
Supplier	Lead Time
Sales	Units Sold, Revenue
Financial	Cost per Unit

- 5 bakery products analyzed
- Excel-based inventory and sales analysis
- Data Source: <https://www.kaggle.com/datasets/preranaaar/bakery-supply-chain-data>

Butter Cookies Recorded the Highest Sales Volume, While Chocolate Cake Showed the Weakest Demand



Key Findings

- Butter Cookies achieved the highest sales volume at 120 units sold
- Chocolate Cake showed the lowest customer demand

Almond Cookies Has the Lowest Inventory Buffer, Increasing Potential Stockout Exposure

Product ID	Product Name	Current Stock	Reorder Point	Inventory Gap
101	Fruit Biscuits	500	200	300
102	Plum Cake	300	100	200
103	Almond Cookies	200	150	50
104	Chocolate Cake	400	250	150
105	Butter Cookies	600	300	300

Key Findings

- Almond Cookies recorded the smallest inventory gap at +50 units
- Butter Cookies and Fruit Biscuits maintain significantly higher stock buffers
- Lower inventory buffer levels increase replenishment urgency

Chocolate Cake Shows Potential Overstock Risk Due to Weak Sales Movement

Product ID	Product Name	Current Stock	Units Sold
101	Fruit Biscuits	500	100
102	Plum Cake	300	50
103	Almond Cookies	200	80
104	Chocolate Cake	400	30
105	Butter Cookies	600	120

Key Findings

- Butter Cookies maintains high inventory levels supported by strong customer demand
- Almond Cookies demonstrates relatively balanced inventory utilization
- Chocolate Cake inventory levels remain high despite lower sales performance

Almond Cookie Presents the Highest Replenishment Risk Due to Extended Supplier Lead Time

Product ID	Product Name	Supplier Lead Time (Days)
101	Fruit Biscuits	7
102	Plum Cake	5
103	Almond Cookies	10
104	Chocolate Cake	7
105	Butter Cookies	5

Key findings

- Almond Cookies has the longest supplier lead time at **10 days**
- Low inventory buffer increases exposure to replenishment delays
- Products with longer lead times require stronger inventory planning

Chocolate Cake Represents the Largest Inventory Investment Across All Products

Product ID	Product Name	Current Stock	Cost per Unit (\$)	Inventory Value (\$)
101	Fruit Biscuits	500	2,5	1250
102	Plum Cake	300	5	1500
103	Almond Cookies	200	3	600
104	Chocolate Cake	400	6	2400
105	Butter Cookies	600	2	1200

Key findings

- Chocolate Cake contributes the highest inventory value due to **higher unit cost**
- Excess inventory exposure may **increase holding cost risk**
- High-value inventory requires tighter inventory control

Inventory Optimization Initiatives Are Required to Improve Operational Efficiency and Reduce Inventory Risk

Recommended Actions

Prioritize Replenishment Planning

- Increase monitoring frequency for Almond Cookies inventory levels
- Prepare earlier replenishment cycles for long lead time products

Reduce Excess Inventory Exposure

- Reevaluate Chocolate Cake Inventory allocation
- Align stock levels more closely with actual demand patterns

Maintain Availability for High-Demand Products

- Sustain sufficient inventory levels for Butter Cookies
- Monitor fast-moving inventory regularly

Executive Summary:

Inventory Optimization Opportunities Were Identified Across Multiple Bakery Products

Summary of Findings

- Almond Cookies presents the highest stockout risk due to low inventory buffer and extended supplier lead time
- Chocolate Cake demonstrates potential overstock exposure driven by weaker sales performance
- Butter Cookies maintains strong sales performance supported by sufficient inventory availability

Strategic Recommendation

Improving replenishment planning and aligning inventory levels with product demand can enhance inventory efficiency and reduce operational risk